Template Document

Academic Support Agent

February 19, 2025

1 About

0.3
160 GB 97M

0.5

3M 13B
Problem Formulation Exact k-NN (Survey bhatia2010survey)
Exact k-NN *Given $(\mathcal{X}, \rho)$ - metric space, $X \subseteq \mathcal{X}$ - set of n points and $q \in \mathcal{X}$ - query point, task of Exact k-NN is to find a set $kNN(q) \subseteq X$ such that $|kNN(q)| = k$ and*

$$\forall x \in kNN(q)\ \forall x' \in X \setminus kNN(q)\ \ (\rho(q, x) \leq \rho(q, x'))$$

Examples of spaces

- $(\mathbb{R}^d, \|\cdot\|_2)$ — Euclidian space
- $\left(\mathbb{R}^d, \arccos \frac{\langle \cdot, \cdot \rangle}{\|\|\cdot\|\|\cdot\|\|}\right)$ — $\mathbb{R}^d$ with cosine distance
- ...

Part I Tools and Systems for Big Data Storage and Processing
enumerate items

1. Hadoop and MapReduce
2. Apache Spark
3. Spark SQL

2

2.1

itemize items

- ($> 10^6$)
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-

3

3.1

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KDDKnowledge Discovery and Data Mining RECSYSRecommender System Conference WWWWorld Wide Web Conferenc